

Tanish Gupta

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Education

Vellore Institute of Technology	India
Bachelor of Technology in Computer Science Engineering	
Specialization in Artificial Intelligence and Machine Learning Science	Aug. 2023 – May 2027

Technical Skills

Languages: Java, Python, JavaScript
Frontend & Backend: React.js, Node.js, Express.js, FastAPI, Django, HTML, Tailwind CSS
Databases: MongoDB, Supabase, MySQL
Tools & Platforms: GitHub, Docker, AWS, Google Cloud, Vercel
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, REST APIs

Projects

Spozo - Sport Venue Booking Platform

Engineered a full-stack sports venue booking platform used to manage 100+ simulated bookings with real-time availability and scheduling.

- Architected scalable full-stack app using React.js, Node.js, and MongoDB, supporting concurrent user requests with <200ms response time.
- Implemented real-time slot booking and authentication system, reducing booking conflicts by ~35%.
- Designed responsive admin dashboard improving booking management efficiency by 40% for venue owners.

AgentOS – Autonomous Multi-Agent Desktop Automation Platform

Architected an autonomous, local-first digital coworker that securely executes and verifies 15+ types of complex desktop workflows across OS applications, running fully offline.

- Built a local-first AI automation platform with 7 specialized agents that autonomously plan, execute, verify, and self-heal complex desktop workflows using Ollama.
- Developed a workflow orchestration engine with 100% local processing, task dependency management, and targeted self-healing, reducing full workflow re-execution by up to 80%.
- Engineered a React + PyWebView dashboard featuring real-time task visualization, execution logs, and live screen monitoring, supporting workflows with 20+ interconnected task nodes.

Automated Irrigation System — AI-Powered Smart Irrigation Platform

Designed an AI-powered irrigation system in Python that predicts crop water needs using soil, weather, and crop data, cutting estimated water usage by ~25%.

- Trained a Scikit-learn based prediction model on 5,000+ soil, weather, and crop-type records to recommend optimal irrigation timing and volume, reducing water wastage by ~25%.
- Implemented a Fast API backend serving real-time predictions in <150ms, paired with a lightweight frontend dashboard for farmers to input 5+ field parameters and view recommendations.
- Used Pandas for data preprocessing and feature engineering across 3 multi-source agricultural datasets, improving model prediction reliability by ~15%.

Hostel Complaint Management System (NLP-Based)

Created a web app-based hostel complaint system enabling real-time complaint registration and tracking, improving issue reporting efficiency by ~40%.

- Integrated secure authentication using Firebase and Google OAuth for 500+ users, ensuring protected access and user-specific complaint tracking.
- Structured backend logic for complaint categorization and automated routing across 5+ departments, cutting manual triage time by ~30%.
- Implemented NLP-based classification to automatically categorize complaints with ~85% accuracy, reducing manual sorting effort.

Certification

Google IT Support Certificate - Google Career Certificates
Machine Learning with Python – IBM (Coursera)
Adobe UI & UX Design - Ethnus
Cloud Computing – IIT Kharagpur (NPTEL)